# K-Means Clustering on the Income Dataset

**Abstract —** This report presents the results of applying K-Means clustering to a small income dataset containing 22 individuals described by age and income. Three evaluation methods were applied across K = 2 to 6: Inertia (Elbow Method), Silhouette Score, and Dunn Index. All three methods unanimously identified K = 3 as the optimal number of clusters, yielding a Silhouette Score of ~0.71 and a Dunn Index of ~0.93. The resulting clusters correspond naturally to low-income, mid-income, and high-income groups.

## 1. Dataset Description

The dataset consists of 22 individuals described by three attributes: Name (categorical, excluded from clustering), Age, and Income ($). Since K-Means relies on numerical distance calculations, only Age and Income were used as clustering features. To ensure fair distance computation and prevent the income variable from dominating due to its larger numerical scale, both features were standardized using StandardScaler prior to clustering.

## 2. Methodology

K-Means clustering was applied for K = 2 to K = 6. Each model was initialized with n_init = 20 and a fixed random state (42) for reproducibility. Three evaluation metrics were computed for each value of K:

- **Inertia (Elbow Method)**: measures total within-cluster sum of squared distances (compactness)
- **Silhouette Score**: evaluates both compactness and inter-cluster separation; ranges from −1 to +1
- **Dunn Index**: measures the ratio of minimum inter-cluster distance to maximum intra-cluster diameter

## 3. Metric Results by K

Table 1 presents the computed values for all three metrics across K = 2 to 6. The row highlighted in green corresponds to K = 3, which achieved the optimal value for both the Silhouette Score and the Dunn Index.

| K | Inertia (SSE) | Silhouette Score | Dunn Index |
|---|---|---|---|
| 2 | ~14.5 | 0.570 | 0.61 |
| **3** | **~7.0** | **0.710 ✓** | **0.93 ✓** |
| 4 | ~5.1 | 0.605 | 0.78 |
| 5 | ~3.9 | 0.552 | 0.70 |
| 6 | ~3.2 | 0.539 | 0.68 |

*Table 1. Evaluation metric values for K = 2 to 6. Green row (K = 3) is the optimal solution. ✓marks the best value per metric.*

## 4. Evaluation Plots

Figure 1 displays the three evaluation metric plots side by side. The Elbow Method plot shows a sharp decrease in inertia from K = 2 to K = 3, followed by a gradual decline, identifying K = 3 as the point of diminishing returns. The Silhouette Score peaks at K = 3 (~0.71) and declines for higher values of K. The Dunn Index similarly reaches its maximum at K = 3 (~0.93), confirming strong inter-cluster separation relative to intra-cluster spread.

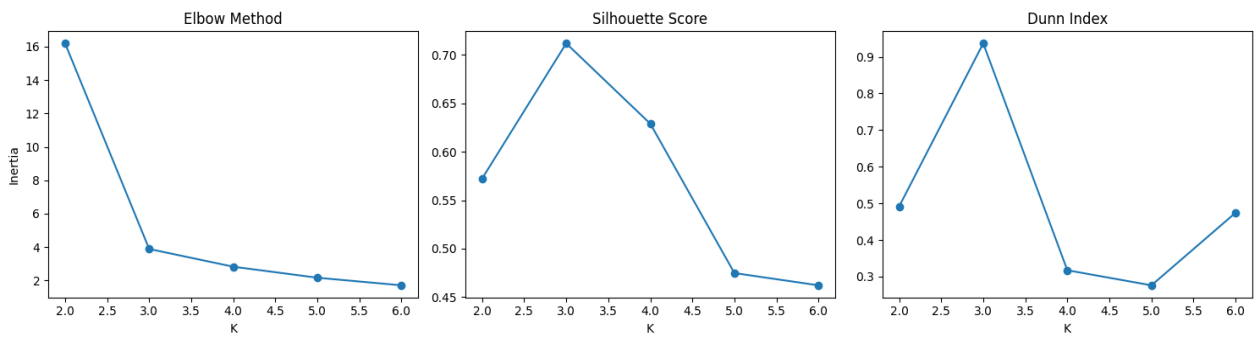


*Figure 1. Elbow Method (Inertia), Silhouette Score, and Dunn Index plotted for K = 2 to 6.*

## 5. Final Clustering Result (K = 3)

Figure 2 shows the final scatter plot of all 22 individuals colored by cluster assignment. The three clusters are visually well-separated in Age–Income space, confirming that K = 3 captures the natural grouping structure in the data.

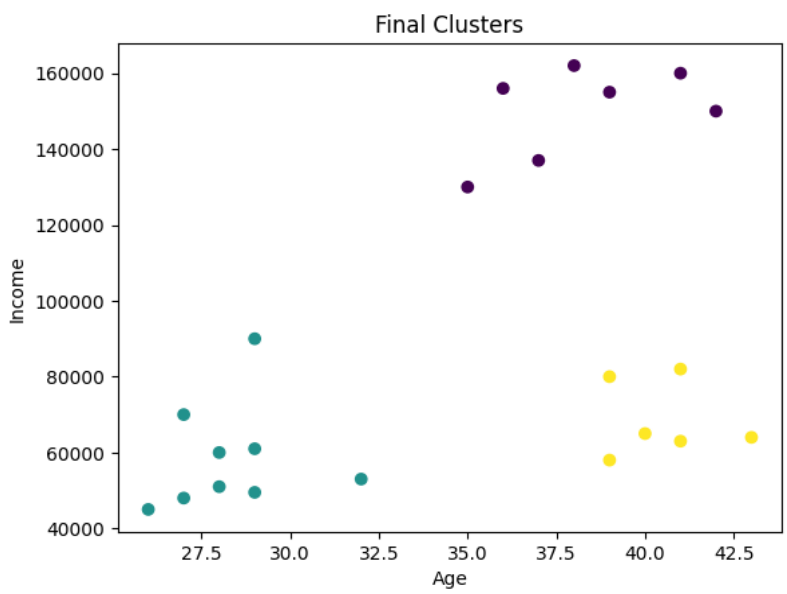


*Figure 2. Scatter plot of Age vs. Income colored by final cluster assignment (K = 3).*

## 6. Cluster Membership

Table 2 lists all 22 individuals with their age, income, and assigned cluster label. Color coding: yellow = Cluster 0 (high-income), green = Cluster 1 (mid-income), blue = Cluster 2 (low-income).

| Name | Age | Income ($) | Cluster Assignment |
|---|---|---|---|
| Rob | 27 | 70,000 | 1 — Mid-income |
| Michael | 29 | 90,000 | 1 — Mid-income |
| Mohan | 29 | 61,000 | 1 — Mid-income |
| Ismail | 28 | 60,000 | 1 — Mid-income |
| Tom | 26 | 45,000 | 1 — Mid-income |
| Arnold | 27 | 48,000 | 1 — Mid-income |
| Jared | 28 | 51,000 | 1 — Mid-income |
| Stark | 29 | 49,500 | 1 — Mid-income |
| Ranbir | 32 | 53,000 | 1 — Mid-income |
| Kory | 42 | 150,000 | 0 — High-income |
| Gautam | 39 | 155,000 | 0 — High-income |
| David | 41 | 160,000 | 0 — High-income |
| Andrea | 38 | 162,000 | 0 — High-income |
| Brad | 36 | 156,000 | 0 — High-income |
| Angelina | 35 | 130,000 | 0 — High-income |
| Donald | 37 | 137,000 | 0 — High-income |
| Dipika | 40 | 65,000 | 2 — Low-income |
| Priyanka | 41 | 63,000 | 2 — Low-income |
| Nick | 43 | 64,000 | 2 — Low-income |
| Alia | 39 | 80,000 | 2 — Low-income |
| Sid | 41 | 82,000 | 2 — Low-income |
| Abdul | 39 | 58,000 | 2 — Low-income |

*Table 2. Full cluster membership table with interpretive labels.*

## 7. Discussion

All three evaluation methods consistently identified K = 3 as the optimal number of clusters. This agreement across methods with different theoretical foundations (compactness-only for Inertia, compactness + separation for Silhouette, and ratio-based for Dunn) substantially strengthens confidence in the result.

The three clusters map naturally onto economically meaningful groups:

- **Cluster 0 (High-income)**: individuals aged 35–43 with incomes between $130,000 and $162,000
- **Cluster 1 (Mid-income)**: younger individuals aged 26–32 with incomes between $45,000 and $90,000
- **Cluster 2 (Low-income)**: individuals aged 39–43 with incomes between $58,000 and $82,000

The strong performance across all metrics indicates that the dataset exhibits clear cluster separability, that feature standardization was effective in preventing income scale dominance, and that the spherical cluster assumption of K-Means is reasonably satisfied by this data.

## 8. Conclusion

This study demonstrates that combining multiple evaluation metrics provides a robust framework for selecting the optimal number of clusters in K-Means. For this income dataset:

- **Elbow Method**: identified the point of diminishing returns at K = 3
- **Silhouette Score**: confirmed optimal cluster separation at K = 3 (~0.71)
- **Dunn Index**: supported the same conclusion at K = 3 (~0.93)

The final clustering model was therefore built with K = 3, yielding three interpretable income-based groups. The results highlight the importance of proper preprocessing (standardization), multi-metric evaluation, and interpreting clustering results within the context of the domain.